Sandhiya Laxmi M

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SUMMARY

Final-year B.Tech student in Artificial Intelligence and Data Science with hands-on experience in machine learning, and data preprocessing. Experienced in developing predictive models, AI-powered applications, and working with real-world datasets using Python. Interested in building intelligent systems and applying AI techniques to practical problem-solving scenarios.

SKILLS

- **Artificial Intelligence & Machine Learning**
- **Programming:** Python, SQL
- **Data Processing:** Pandas, NumPy, PySpark
- **Web Basics:** HTML, CSS, JavaScript
- **Tools:** Git, GitHub, VS Code, Vercel

PROJECTS

Machine Learning Framework for Student Dropout analysis and counselling
Tech Stack: *Python, Scikit-learn, Pandas, Groq API*
- Developed a machine learning system to predict student dropout risk using academic and behavioral data.
- Performed data preprocessing including feature encoding, dataset cleaning, and preparation for model training.
- Implemented a Random Forest model to identify high-risk students.
- Integrated an AI-powered counseling assistant using the Groq API to provide interactive chat-based guidance.

Image Captioning Model (CNN + GRU with Attention)
Tech Stack: *TensorFlow, InceptionResNetV2*
- Developed a deep learning model capable of generating natural language captions from images.
- Used CNN-based feature extraction and GRU networks for sequence modeling.
- Implemented an attention mechanism to improve caption accuracy and contextual understanding.

Autocleanr-Scale – Distributed Data Processing Pipeline
Tech Stack: *PySpark, Python, Pandas, Parquet*
- Developed a scalable data validation and preprocessing pipeline for large structured datasets.
- Designed ETL workflows for cleaning, transformation, and schema standardization.
- Optimized distributed execution using PySpark for improved performance.

EXPERIENCE

Machine Learning Intern | Vaidsys
- Performed data cleaning, validation, and structured data preparation for model training.
- Assisted in developing small ML workflows and evaluating model performance.

ACHIEVEMENTS

- **Winner – Idea Pitching Competition (NABARD MABIF)** – Awarded a cash prize of ₹25,000
- **First Place – Masters Query Madness (SQL)** – TCE College – Awarded a cash prize of ₹2,000
- **Toastmasters International – Humorous Speech Contest:** Club Level: 2nd Place | Area Level: 3rd Place

EDUCATION

B.Tech – Artificial Intelligence and Data Science | 2022- 2026
Sethu Institute of Technology – Tamil Nadu | CGPA: **8.45 / 10**

CERTIFICATIONS & BADGES

- **Generative AI and LLMs: Architecture and Data Preparation** – Skill Badge (Google Cloud)
- **Python for Data Science, AI & Development** – Coursera
- **Prompt Design in Vertex AI** – Google Cloud